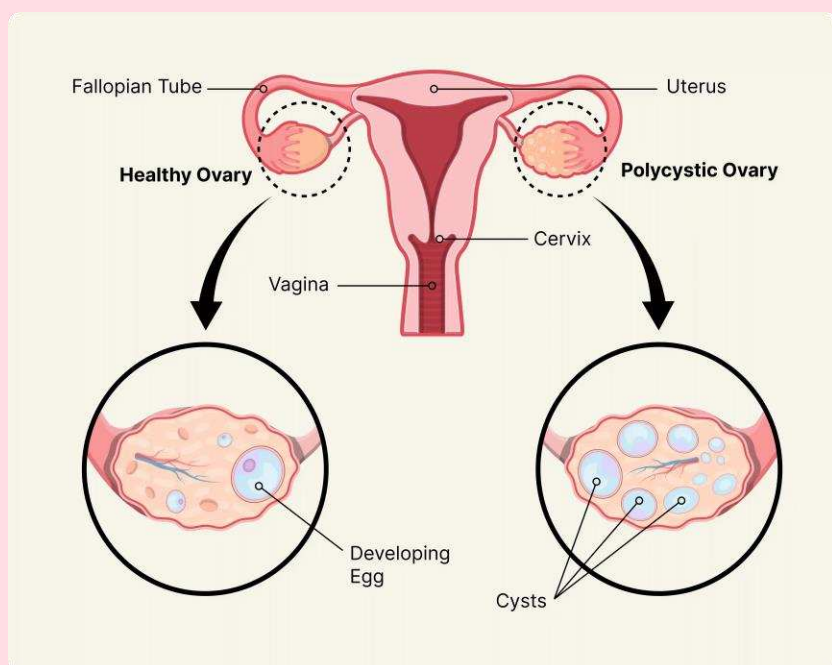


Hypothesis

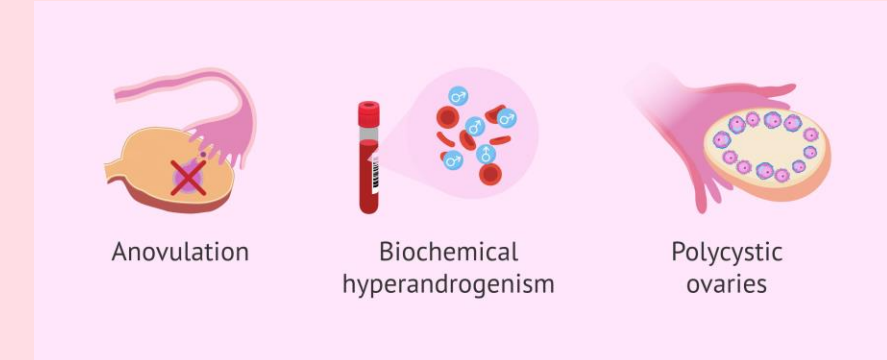
If *Drosophila melanogaster* are induced with PCOS and then fed natural antiandrogens like spearmint, then the severity of their PCOS symptoms will be reduced because antiandrogens treat hyperandrogenism which is associated with PCOS.

Introduction

Polycystic Ovary Syndrome (PCOS) is a hormonal disorder affecting 8–13% of reproductive-aged women, with about 70% remaining undiagnosed.



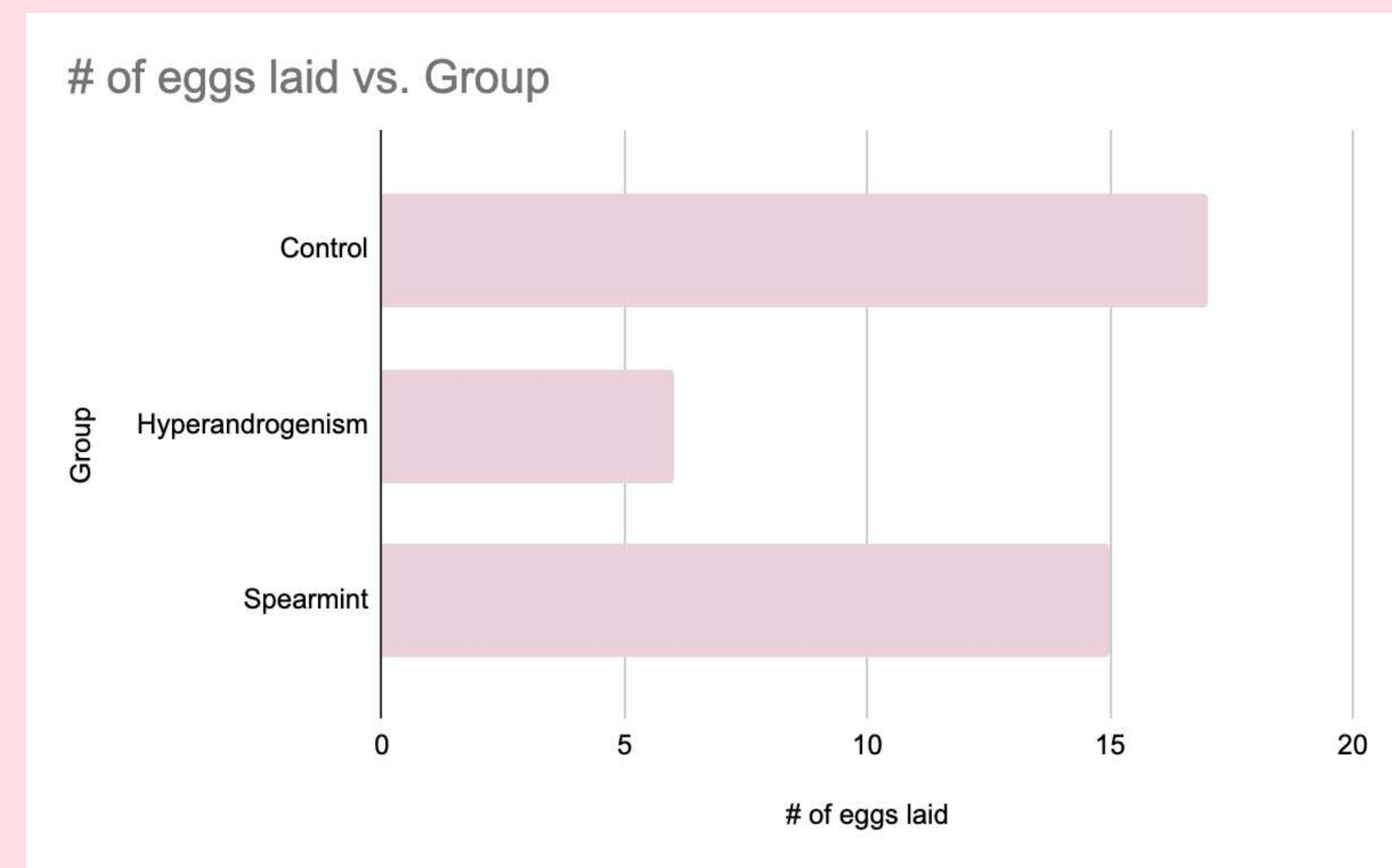
Hyperandrogenism, a key trait of PCOS, is treated with antiandrogens that block hormone receptors in target tissues. **Spearmint**, a natural antiandrogen, has been shown to help lower androgen levels and improve hormonal balance, making it a promising option for symptom management.



20 hydroxy ecdysone is the “active” form of ecdysone. Ecdysone is similar to human testosterone, and 20-hydroxyecdysone is similar to human 20-hydroxytestosterone (Kilpatrick, n.d.).

Graphs

Group	Avg. # eggs laid
Control	17
Hyperandrogenism	6
Spearmint	15



The Effect of the Natural Anti-Androgen Spearmint on *Drosophila melanogaster* Induced with Polycystic Ovary Syndrome (PCOS)-like symptom: Hyperandrogenism

Methods

Inducing PCOS symptom: Hyperandrogenism

- 1 Make starvation media by mixing 1 gram of agar into 100mL of boiling water
- 2 Micropipette 500µL of absolute ethanol into 5 mg of 20E
- 3 Micropipette 20µL of 20E and absolute ethanol solution onto surface of starvation media
- 4 Let the vials sit for 24 hours to allow the ethanol to evaporate

Setting up Apple Juice Agar Plates for Egg Laying Media

- 1 Mix 4.5 grams of agar in 150ml of H₂O
- 2 Mix 5 grams of sucrose in 50 ml apple juice
- 3 Combine both solutions and stir
- 4 Once cooled to 60°C, add propionic acid and pour solution into plates

Administering Natural Antiandrogen (Spearmint)

- 1 Combine 1.95 grams of spearmint powder into 195 mL of boiling water
- 2 Mix solution into the dry ingredients and corn syrup for a half batch of fly food
- 3 Pour fly food into vials and allow to cool
- 4 Create 3 vials of 20 Females and 10 males. When administering a treatment, vial contains only 20 Females

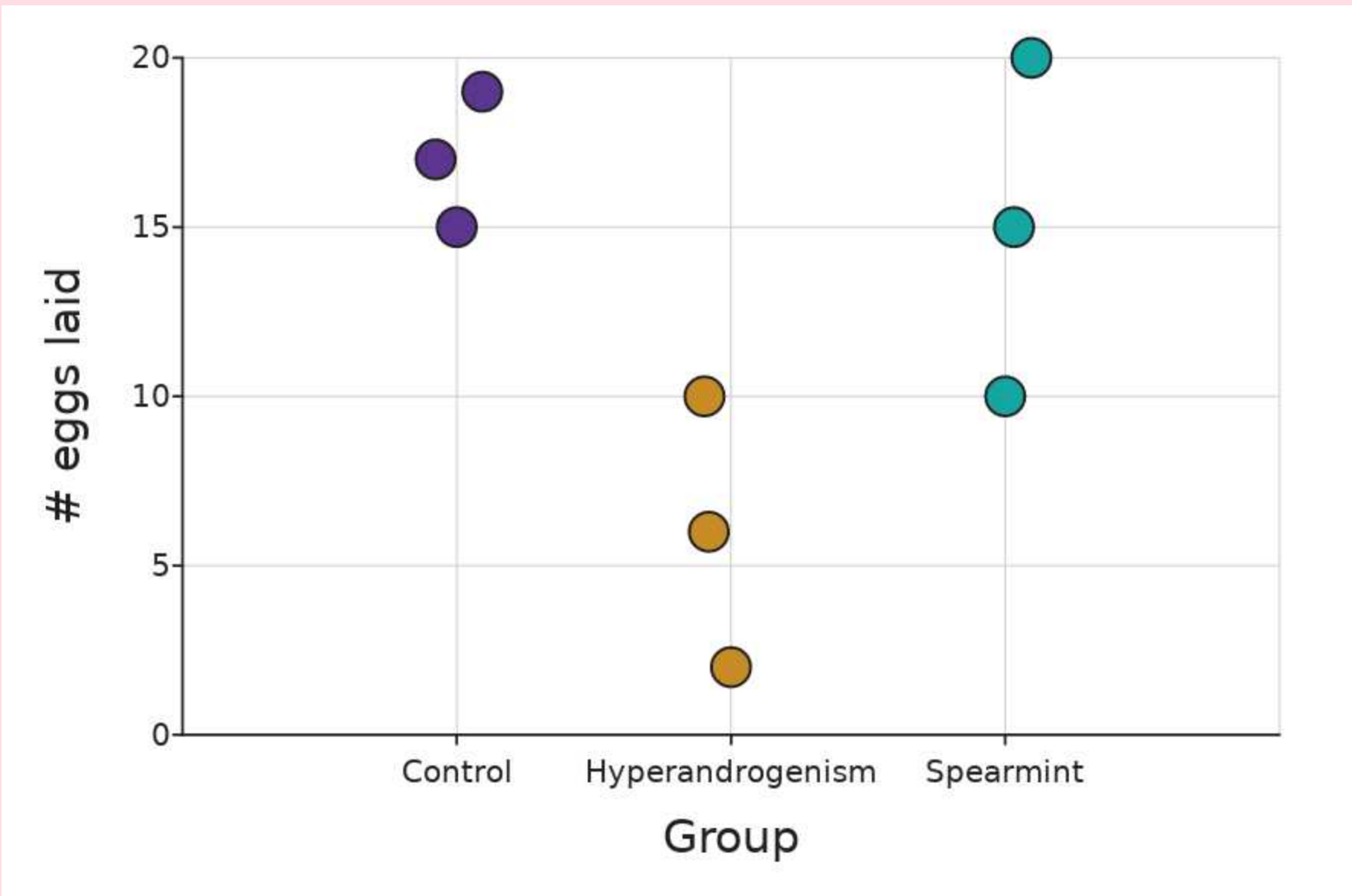
Data Collection

- 1 Cold Sort
- 2 Tap *Drosophila* onto experimental media and start a 24 hr clock
- 3 Use microscope to count eggs and larvae

Results

The reproductive output of *Drosophila* across the three experimental groups was evaluated using a Kruskal-Wallis test. While the control and spearmint groups showed higher mean egg-laying rates relative to the reduced output of the hyperandrogenism group, the overall variation among the group medians did not achieve strict statistical significance ($H(2) = 5.04$, $p = 0.08$). Consequently, a post-hoc Dunn's test with Bonferroni correction revealed no statistically significant differences in pairwise comparisons, including Control vs. Hyperandrogenism ($p = 0.11$) and Hyperandrogenism vs. Spearmint ($p = 0.25$).

Degrees of Freedom (df)	H-statistic	P-value	Interpretation of P
2	5.04	0.08	A P-value of 0.08 means some confidence that the groups are different.



Discussion

Although the descriptive data indicates a clear biological trend, where induced hyperandrogenism suppresses egg production and spearmint intervention appears to restore it, the Kruskal-Wallis test did not detect a statistically significant difference ($p = 0.08$). This marginal significance is highly likely an artifact of low statistical power resulting from the small sample size ($n = 3$ replicates per group), which inflates the impact of intra-group variance. While these preliminary findings support the hypothesis that spearmint may counteract androgen-induced reproductive dysfunction via its known anti-androgenic properties, further replication with larger sample sizes is required to definitively confirm these statistical trends.

Conclusion

While induced hyperandrogenism visually suppressed *Drosophila* egg production and spearmint intervention trended toward rescuing this phenotype, these differences were not statistically significant due to small sample limitations ($p=0.08$). Future studies with larger sample sizes are required to definitively confirm spearmint’s therapeutic potential in mitigating androgen-induced reproductive dysfunction.

Future Work

Future work will measure the effects of other natural anti-androgens to determine which is most effective in raising the fecundity of the PCOS-induced *Drosophila melanogaster* as well as utilizing larger sample sizes. These natural anti-androgens include, but aren't limited to, flaxseed and green tea. Additionally, more research can be conducted on the effects of spearmint on reproduction in wild type *D. melanogaster*, and a more effective method for inducing flies with PCOS can be found.

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